HIGH PRESSURE PROCESS WATER PUMPS
2000-F17-PID-PR-10002
HP PROCESS WATER
17-0174-PW-PE16-160

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0360-PW-PE8-225
PE8 SS1
17-0361-PW-SS1-200
200VBFD1S2 F17-10837
200VBFD1S1 F17-10838
50VB03S1 F17-10841
200x150
150VBFD1S1 FC
200x150
200VBFD1S1 F17-10839
200VCD1S1 F17-10842
300x200
TO GRADE
17-0925-PW-SS3-50
200VBFD1S1 F17-10840

FLOCCULANT DOSING PUMPS
2000-F72-PID-PR-10004
FLOCCULANT
17-0371-FL-PE8-90
PE8 MS1
17-0372-FL-MS1-80
80VBD1C1 F17-10843
80VBD1C1 F17-10844
25VB03S1 F17-10845
17-0275-FL-MS1-25
TO GRADE

FLOTATION AREA SUMP PUMP 2
2000-F16-PID-PE10-160
SPILLAGE
16-1208-FD-PE10-160

FLOTATION AREA SUMP PUMP 3
2000-F16-PID-PR-10004
SPILLAGE
16-1214-FD-PE10-160

CLEANER AREA SUMP PUMP
2000-F18-PID-PR-10001
SPILLAGE
18-1216-FD-PE10-110

FLOCCULANT AREA SUMP PUMP
2000-F72-PID-PR-10004
SPILLAGE
72-1076-FL-PE8-90

FLOTATION TAILINGS THICKENER UNDERFLOW PUMP
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0383-CON-PE10-500

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0187-PW-PE8-500
500VBFD1L2 F17-10846

FLOTATION CELLS DRAIN PUMP
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1290-FS-PE10-280

FLOTATION AREA SUMP PUMP 1
2000-F16-PID-PR-10001
SPILLAGE
16-1206-FD-PE10-160

RAW WATER PUMPS
2000-F24-PID-PR-10002
RAW WATER
17-0300-BW-PE8-63
PE8 SS3
17-0975-BW-SS3-50
50VC01S1 F17-10847
17-1424-BW-SS3-25
25VB03S1 F17-14149
50x25
25VB03S1 F17-10848
25VB03S1 F17-10849
25VB03S1 F17-14109
SP MC-001
SP HS-025
F17-XM-572 FLOTATION TAILINGS THICKENER MANTA SUB
1/2" BSP FEMALE
DETAIL 5
HS 17009
SP HS-056
SP HS-056
SP SC-034
XV 17009
ID25mm ID25mm
DN1200 DN1200

FLOTATION DISCHARGE BOX
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1248-FS-RL1-1200
1.5°
F17-SA-416 FLOTATION TAILINGS THICKENER FEED SAMPLER
VENDOR PACKAGE 2000-ME074-J02-10007
DN50
17-1742-FS-RL1-1200
17-1743-FS-RL1-80
80VKGD1L1 F17-10851
1200VKGD1L2 FL
SP HS-105
SP HS-105
TO GRADE
F17-HU-422 FLOTATION TAILINGS THICKENER FEED BOX VALVE HYDRAULIC POWER UNIT
17-0497-FS-RL1-1200
TI 17141
PI 17140
LI 17142
F17-BX-410 FLOTATION TAILINGS THICKENER RECIRCULATION FEED BOX
THICKENER FEED STANDPIPE NOTE 6
F17-PP-459 FLOTATION TAILINGS P-DUC PUMP
F17-TH-401 FLOTATION TAILINGS THICKENER
F17-XM-474 F17-XM-471 F17-XM-613 F17-XM-472 F17-XM-614 F17-XM-473
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER DART VALVES (6 OFF)

PE10 RL1
17-0368-FD-PE10-225
17-0367-FD-RL1-200
50VB03S1 F17-10853
SP MC-002
LAH 17081
LSH 17081
200VKGD1L1 F17-10854
50VB03S1 F17-10852
SP MC-002
200x100
NOTE 7
F17-PS-427 FLOTATION TAILINGS THICKENER AREA SUMP PUMP 1

16-0496-FS-PE10-63
LIC 17004
LT 17004
LI 17004
DETAIL 3
HS 17005
XV 17005
80VKGD1L1 FL
16-3072-FS-PE10-110
TO GRADE
F16-HP-456 FLOTATION TAILS SAMPLE HOPPER
16-1515-FS-SS3-50
TO GRADE
ZAH 17144
PAH 17143
(6 OFF)
SP MC-001
25VB03S1 F16-10855
16-0660-FS-RL1-80
16-2140-FS-RL1-80
80x65
80x65
DETAIL 4
HS 17006
XV 17006
50VKGD1L1 F0 RL1 SS3
SP MC-001
25VB03S1 F17-10856
DETAIL 4
HS 17007
XV 17007
50VKGD1L1 F0 RL1 SS3
16-1448-FS-SS3-50
TO GRADE
RL1 PE10
90x63
BURST
XA 17008
XS 17008
SP HS-074
SP HS-074
F16-PP-634 FINAL TAILS SAMPLE PUMP

17-0373-FL-SS1-300
F17-XM-413 FLOTATION TAILINGS THICKENER FLOCCULANT INLINE MIXER
17-0130-PW-SS3-50
PE16 SS3
17-0267-PW-PE16-63
SP HS-100
25VB03S1 F17-10821
50VB03S1 F17-10822
SP MC-001
SP MC-002
SP @ BRIDGE OF THICKENER F17-TH-401
CCTV F17-CAM-001
SP SC-034
NOTE 5
VSD
DN150
TAH 17039
AIT 17019
AE 17019
AI 17019
ZS 17010
TURBIDITY
LIT 17012
LE 17012
BED LEVEL METER MANTA SUB
RAKE HEIGHT LO
RAKE HEIGHT HI
RAKE UNDERSPEED SWITCH
GEARBOX LEVEL SWITCH LOW
GEARBOX LEVEL SWITCH LOW
RAKE HEIGHT POSITION
ZSL 17015
ZSH 17014
SSL 17016
LSL 17018
LSL 17017
ZIT 17013
ZE 17013
F17-XM-572 MANTA SUB
F17-HU-406 FLOTATION TAILINGS THICKENER RAKE DRIVE HPU
F17-HU-414 FLOTATION TAILINGS THICKENER RAKE LIFT HPU
VENDOR PACKAGE 2000-ME0015-E02-50001 TO 50006
F17-VL-458 FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER
F17-BX-464 FLOTATION TAILINGS THICKENER OVERFLOW BOX
F17-XM-572 FLOTATION TAILINGS THICKENER MANTA SUB
SP SC-034
SP SC-034
F17-VL-490 FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 1
F17-VL-491 FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 2
BED PRESSURE PIC 17011 HH H L
PE 17011 BED MASS METER
DN50 50VB03S1 F17-10857
SP MC-002
SP MC-002
50VB03S1 F17-10858
SP MC-002
17-0498-CON-RL1-300
F17-10863 300VKGD1L1
DN50 50VB03S1 F17-10859
DN50 50VB03S1 F17-10860
DN50 50VB03S1 F17-10861
DN50 50VB03S1 F17-10862
TO GRADE
1:100
17-0364-CON-RL1-600
17-0374-CON-RL1-600
17-0365-CON-RL1-600
17-0377-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1 2000-F17-PID-PR-10002 FLOTATION TAILINGS THICKENER UNDERFLOW
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2 2000-F17-PID-PR-10002 FLOTATION TAILINGS THICKENER UNDERFLOW
16-0661-FS-PE10-63
FLOTATION PARTICLE SIZE ANALYSER 2000-F16-PID-PR-10005 FINAL TAILS SAMPLE

17-0982-PW-PE16-160
17-0983-PW-PE16-160
PE16 SS2 SS3
17-0976-PW-SS2-100 100VBFE1S1 F17-10824 17-1487-PW-SS3-50 50VB03S1 F17-10825
17-0984-PW-SS2-100 100VBFE1S1 F17-10826 17-1486-PW-SS3-50 50VB03S1 F17-10827
17-0985-PW-SS2-100 100VBFE1S1 F17-10828 17-1485-PW-SS3-50 50VB03S1 F17-10829
17-0986-PW-SS2-100 100VBFE1S1 F17-10830 17-1484-PW-SS3-50 50VB03S1 F17-10831
17-0987-PW-SS2-100 100VBFE1S1 F17-10832 17-1483-PW-SS3-50 50VB03S1 F17-10833
17-0988-PW-SS2-100 100VBFE1S1 F17-10834 17-1482-PW-SS3-50 50VB03S1 F17-10835
F24-XM-451 F24-XM-452 F24-XM-453 F24-XM-454 F24-XM-455 F24-XM-456
FLOTATION TAILINGS THICKENER WATER MONITORS (6 OFF) NOTE 7
SP MC-002

PI 17002 PI 17002 PT 17002 50VBE1S1 F17-10823
FI 17003
FIT 17001 FE 17001
DETAIL 1 FIC 17001 H L
FCV 17001
f(x) DILUTION FACTOR
FIT 17003 FE 17003
MANTA SUB START/STOP HS 17038
MANTA SUB RUNNING XI 17038A
MANTA SUB FAULT XA 17038B
MANTA SENSOR FAILED TO RETURN HOME ALARM XA 17038
NOTE 4
XA 17060 EMERGENCY STOP
ZAH 17014 MAX. RAKE LEVEL
ZAL 17015 MIN. RAKE LEVEL
XI 17066 24VDC POWER HEALTHY
SV 17033 RAKE DRIVE FORWARD SOLENOID RELAY
SV 17034 RAKE DRIVE REVERSE SOLENOID RELAY
SV 17048 RAKE LIFT RAISE SOLENOID RELAY
SV 17047 RAKE LIFT LOWER SOLENOID RELAY
XC 17076 RAKE DRIVE RUNNING FEEDBACK
XC 17057 LOCAL MODE ENABLE FEEDBACK
NI 17030 TORQUE INDICATION
XC 17079 85% TORQUE FROM DCS TO LCP
XC 17083 50% TORQUE FROM DCS TO LCP
XC 17080 40% TORQUE FROM DCS TO LCP
XC 17084 30% TORQUE FROM DCS TO LCP
HS 17065A RAKE LIFT LOCAL RAISE REQUEST
HS 17065B RAKE LIFT LOCAL LOWER REQUEST
HS 17063 REVERSE JOG LOCAL START REQUEST
XC 17082 ALARM ACKNOWLEDGE FROM DCS TO LCP
XC 17062 REVERSE JOG ACKNOWLEDGE FROM DCS TO LCP
HS 17061 RAKE DRIVE FORWARD LOCAL START REQUEST
HS 17064 RAKE DRIVE LOCAL STOP REQUEST
PDAH 17036 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (WORKING)
PDAH 17028 RAKE DRIVE OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
PDAH 17037 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (STANDBY)
PDAH 17031 RAKE DRIVE OIL TANK OUTLET OIL DIFF PRESSURE
ZI 17013 RAKE LIFT POSITION
TAH 17022 RAKE DRIVE OIL TANK TEMPERATURE HIGH
2000-F72-PID-PR-10004 SIGNAL
FIC17003
TAHH 17024 RAKE DRIVE OIL TANK TEMPERATURE HIGH
LALL 17023 RAKE DRIVE OIL TANK LEVEL LOW LOW
LAL 17021 RAKE DRIVE OIL TANK LEVEL LOW
LAL 17041 RAKE LIFT OIL TANK LEVEL LOW
TAH 17042 RAKE LIFT OIL TANK TEMPERATURE HIGH
LALL 17043 RAKE LIFT OIL TANK LEVEL LOW LOW
TAHH 17044 RAKE LIFT OIL TANK TEMPERATURE HIGH HIGH
PDAH 17051 RAKE LIFT OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
LAL 17017 GEAR REDUCER OIL LEVEL LOW
LAL 17018 GEAR REDUCER OIL LEVEL LOW
SAL 17016 RAKE ZERO SPEED
ZAO 17026 F17-PO-408 THICKENER RAKE DRIVE PUMP SUCTION VALVE POSITION SWITCH OPENED
PROCESS WATER TANK 2000-F24-PID-PR-10001 FLOTATION TAILINGS THICKENER OVERFLOW
PROCESS WATER TANK 2000-F24-PID-PR-10001 FLOTATION TAILINGS THICKENER OVERFLOW
FIC17121 2000-F17-PID-PR-10002 SIGNAL

NOTES:
1. REFER TO CONTROL PHILOSOPHY FOR COMPREHENSIVE LIST OF PROCESS INTERLOCKS AND CONTROL FUNCTIONS.
2. STAGE 2.
3. TIE IN POINT LOCATED ON F17-BX-410.
4. SIGNAL FROM FLOTATION THICKENER LCP TO MCC/DCS.
5. FLOCCULANT IS BEING DISTRIBUTED WITHIN VENDOR PACKAGE.
6. THERE ARE 3 OFF SPARE NOZZLES (DN100, DN200 & DN60 AVAILABLE).
7. FLANGED ASME B16.5 CLASS 150.

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	12SEP25	LAUNDERS AND GENERAL REVISIONS ADDED					DM
1	19NOV24	RE-ISSUED FOR CONSTRUCTION					
0	06SEP24	APPROVED FOR CONSTRUCTION					

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FLOTATION TAILINGS PRE-LEACH THICKENING
FLOTATION TAILINGS THICKENING - SHEET 1 - STAGE 1
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F17-PID-PR-10001	2

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\PR\PID\ISSUED\12.09.25 - SIGNING\NATIVES\2000-F17-PID-PR-10001_2.DWG 10SEP25 10:19AM ANNE.HOYNE